YEAR 10 MATHS · CHAPTER 1 · MIND MAP

Algebra, equations and linear relationships

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1 Review of algebra

Collect like terms; expand by multiplying every term; factorise by taking out the HCF.

$5a - 2b + 3a = 8a - 2b$

$3x(2 - x) = 6x - 3x^2$

$6ab - 9a = 3a(2b - 3)$

2 Solving linear equations

Undo each operation in reverse order, doing the same to both sides.

$\frac{2x+1}{3} = 5 \Rightarrow 2x + 1 = 15 \Rightarrow x = 7$

$5(x - 1) = 2x + 10 \Rightarrow 3x = 15 \Rightarrow x = 5$

3 Linear inequalities

Solve like an equation, but flip the sign when you multiply or divide by a negative.

$3 - 2x \leq 11 \Rightarrow -2x \leq 8 \Rightarrow x \geq -4$

Watch out: flip the sign, not just the number.

4 Equations with algebraic fractions

Multiply every term by the lowest common denominator first.

$\frac{x}{3} + \frac{x-1}{2} = 4$: $\times 6$, $2x + 3(x - 1) = 24$, $x = 5.4$

$\frac{x+2}{4} = \frac{x-1}{3} \Rightarrow 3x + 6 = 4x - 4 \Rightarrow x = 10$

5 Graphing straight lines

$y = mx + c$. Or find the two intercepts and join them.

$3x - 2y = 6$: x-intercept $(2, 0)$, y-intercept $(0, -3)$

$y = -\frac{1}{2}x + 4$: gradient $-\frac{1}{2}$, y-intercept 4

6 Finding the equation of a line

Find the gradient, then use $y - y_1 = m(x - x_1)$.

Through $(2, 3)$ and $(4, 11)$: $m = \frac{8}{2} = 4$

$y - 3 = 4(x - 2) \Rightarrow y = 4x - 5$

7 Length and midpoint

Midpoint: average the coordinates. Length: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$(-1, 2)$ and $(5, 10)$: midpoint $(2, 6)$

Length $\sqrt{6^2 + 8^2} = 10$

8 Parallel and perpendicular lines

Parallel: equal gradients. Perpendicular: $m_1 \times m_2 = -1$.

Perpendicular to $y = 3x + 1$: gradient $-\frac{1}{3}$

Through $(3, 2)$: $y = -\frac{1}{3}x + 3$

9 Simultaneous equations: substitution

Replace one pronumeral with what it equals from the other equation.

$y = 2x - 1$ and $3x + y = 14$

$3x + 2x - 1 = 14 \Rightarrow x = 3,\ y = 5$

10 Simultaneous equations: elimination

Add or subtract the equations so one pronumeral cancels.

$2x + 3y = 12$ and $4x - 3y = 6$

Add: $6x = 18$, so $x = 3,\ y = 2$

11 Applications of simultaneous equations

Define two pronumerals, write two equations, solve, then answer in words.

Two numbers add to 31 and differ by 7: $x + y = 31,\ x - y = 7$

Add: 2x = 38, so the numbers are **19** and **12**

12 Regions on the Cartesian plane

Draw the boundary line (dashed for $<$ or $>$, solid for $\leq$ or $\geq$), then test a point.

$y > 2x + 1$: test $(0, 0)$: $0 > 1$ is false

So shade the side without the origin